CS556 Fall 2022 Test 1

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes. *Read carefully before answering!* Write your answers on this test paper. Also hand in your note sheet, with your name on it.

Question 1 *(10 points)* Let $A = \begin{bmatrix} -2 & 3 \\ 2 & 3 \end{bmatrix}$

(a) Find eigenvalues and an independent set of eigenvectors.

(b) Diagonalize A.

Question 2 *(5 points)* Assume **u** and **v** are unit vectors. What is the value of the dot product $(\mathbf{u} + \mathbf{v})\cdot(\mathbf{u}-\mathbf{v})$?

(a) 2

(b) 1

(c) 0

(d) None of the above, we need to know the vectors **u** and **v**.

Question 3 *(5 points)* The two vectors $(-1, 1-a^2, 1)$ and $(-1, 1, a^2)$ are orthogonal for all values of a:

(a) True

(b) False

Question 4 *(5 points)* If A and B are $n \times n$ symmetric matrices, that is, $A^T = A$ and $B^T = B$, then ABA is symmetric matrix.

(a) True

(b) False

Question 5 *(10 points)* Use Gauss-Jordan elimination to find A^{-1} if

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

Question 6 *(15 points)* Consider the following system of linear equation for some constants b and g:

$$1x + by + 0z = 1$$
$$1x - 2y - 1z = 2$$
$$0x + 1y + 1z = g$$

a) Which number b leads to a row exchange?

b) Which number b makes the system singular (missing a third pivot)?

c) For the b found in part (b), for which value of g, the system has infinitely many solutions?

d)Find the solutions of the system with b and g found in parts (b) and (c), if z = 2

Question 7 *(10 points)* Construct a matrix with the required property or explain why that is impossible.

(a) Column space contains $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Question 8 *(10 points)* Find the determinant of matrix A:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & -1 & -2 \end{bmatrix}$$

Question 9 *(15 points)* Find a basis for each of the four fundamental subspaces of the matrix.

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Question 10 *(15 points)* Let $a_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

(a) Find the projection of the vector $b = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ onto the subspace spanned by a_1 and a_2.

(b) Find the error vector and its magnitude.

(c) Use Gram-Schmidt process to find an orthonormal basis for the span of a_1 and a_2.